Proof-Carrying Detection of Reticulate Evolution: a Tree-Level Null, the Measured Cost of the Alignment Step, and the Gromov Hinge of Buneman's Converse

Xavier Callens

SocrateAI Scientific Research & Formal Verification Laboratory

Formal artifacts: `ANSE.GeneticaReticulata`, `ANSE.GeneticaReticulataTree` (Lean 4)

Edition 5, 23 October 2026. Identifies the certificate's margin with a Bandelt–Dress isolation-index gap and proves it (§14), which replaces edition 4's closure of the topological line with a positive statement about what the certificate was already measuring. **Not peer reviewed.**

Abstract

Tests for recombination are normally calibrated against a permutation ensemble. We develop one calibrated against a *theorem*, report the two ways our own first version of it was wrong, and give the measurement that we believe is new.

A strict molecular clock on a clonal genealogy makes expected p-distances ultrametric, and an ultrametric admits no unfilled Vietoris–Rips cycle at any scale; three sequences violating the strong triangle inequality are therefore a *certificate*, not a surprise relative to shuffles. We formalise the inference rule in Lean 4 and emit certificates that embed the aligned rows, so the kernel recomputes every count by `rfl` rather than trusting a search.

The statistic this licenses is nonetheless the wrong one, and a control we had not run says so. On a clonal tree with a **relaxed clock and no reticulation whatever**, divergence-matched so that only the clock differs, the ultrametric margin exceeds every single-recombinant margin in the positive control; at 32 leaves $\min(\text{no reticulation}) = 99/4000 > 79/4000 = \max(\text{one recombinant})$ at every seed. The four-point condition encodes *tree* without *clock*, we prove in Lean that every symmetric ultrametric satisfies it, and the quartet margin on the same data does not move. So the quartet statistic must carry the method, and the licence for reading the gap between the two conditions is a theorem rather than a convention.

Running this on real data required writing the alignment step, which puts an aligner on the trust surface that no downstream proof covers. We measure it instead of caveating it: on simulated populations with no reticulation and a known true alignment, reference-anchored alignment manufactures 0 of 9100 four-point violations at a 0.4% gap load and 1600 of 9100 at 8.2%. We find no published benchmark that varies alignment quality as an independent variable, so we believe this is the first direct measurement of a mechanism the field already asserts. Read as an admissibility threshold it refuses 7 of 10 real viral populations drawn from a public corpus; the 3 that pass agree with a literature prior fixed in advance, and the two certificates clearing their matched floor are accepted by the kernel with both corrupted twins rejected.

Two corrections to our own apparatus are reported as results. The band-edge test by which we had certified the aligner is not a soundness certificate, and it had really failed: a pair scoring 16555 against an unbanded optimum of 17275 with the band edge never touched. Certifying every alignment against the unbanded optimum dropped the figure above from 4876 to 1600 — two thirds of our headline was an uncertified step inside the measuring apparatus. Separately, the statistic we report is a maximum over subsets where the established treelikeness tests are averages; run seed-paired against the hard null, the

maximum separates 16.57× against 5.07× for mean δ_q and 1.22× for the proportion of violating quadruples.

Edition 1 listed the missing head-to-head as its chief limitation. We now run it against the one incumbent whose null also assumes no clock, the pairwise homoplasy index, on the same alignments under the same deletion policy, and the result revises what this method may claim. **On the five populations where both tests can be measured the two agree exactly**: both are specific on the strict-clock *and* the relaxed-clock null, and both detect all three reticulate populations at every seed. The incumbent is not beaten on power, so the case for a certificate cannot rest on power. It rests instead on two things the comparison does not touch — an exhibitable witness a kernel recomputes, and the absence of a permutation null, which matters because the artefact class we measure in Section 5 is reported to be invisible to exactly that kind of null.

Finally we prove the hinge of Buneman's converse, which this programme had only cited: a symmetric dissimilarity satisfies the four-point condition if and only if its Gromov product satisfies the max inequality at every root. We find no formalisation of tree metrics, splits or Buneman's theorem in Lean, Isabelle or Coq, so we believe this is first of its kind. All Lean declarations compile with zero `sorry` and no axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}.

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1 Introduction

A test for recombination has to answer the question: compared with what? The standard answer is an ensemble — shuffle the columns of the alignment, recompute the statistic, and read off a p-value. It is a serviceable answer with two costs. The first is a resolution floor: with N shuffles no p-value below $1/(N+1)$ is attainable, so the apparatus has an opinion about what counts as significant before the data arrives. The second is sharper and less often stated. The manual of the field's dominant tool observes that misalignment produces false recombination signals which *"will be detected as recombination events with significant p-values"* while *"these false positive signals will likely be undetectable in the shuffled alignments"* [10]. Column shuffling destroys the spatial clustering of misalignment along with the signal, so an entire class of artefact survives the real analysis and vanishes from the null. A permutation null cannot see it.

This paper develops the alternative: a null that is a theorem, so that a violation is a witness rather than a tail probability.

Modelling assumption 1.1 (What the clock-level null assumes)**.** Write $d(x, y)$ for the p-distance between two aligned sequences — differing columns over columns comparable in both. Expected p-distances satisfy

$$d(x, z) \;\le\; \max\big(d(x, y),\, d(y, z)\big) \qquad \text{for all } x, y, z \tag{1}$$

exactly when descent is **clonal** *and* the molecular clock is **strict**. Two assumptions, one inequality.

If (1) holds then $d(\cdot,\cdot) \le \epsilon$ is transitive, so every connected component of the Vietoris–Rips complex at scale ϵ is a clique and no unfilled cycle exists at any scale. Three sequences with $\max(d(x, y), d(y, z)) < d(x, z)$ therefore contradict a universally quantified statement. Nothing is calibrated.

Section 2 states the two nulls and proves they nest. Section 3 refutes the clock-level statistic with a control that contains no reticulation. Section 4 deals with missing data. Section 5 is the paper's main empirical contribution: the alignment step is on the trust surface, and this is how large it is. Section 6 reports the proxy inside our own apparatus that failed and what it had inflated. Section 7 prices the maximum against the average, and Section 8 runs the comparison against the incumbent that edition 1 was missing. Section 12 runs the test on real viral populations. Section 13 proves the hinge of Buneman's converse. Section 15 says what would make all of it wrong.

What is and is not claimed. We claim: the nesting theorem and the Gromov equivalence, both machine-checked; the refutation of the clock-level statistic on simulated data; the measurement of alignment-manufactured violations; and three real-data verdicts that agree with priors fixed in advance. We do not claim a validated method. Three admissible populations is not a validation, no head-to-head against the established tests has been run, and Buneman's converse remains cited for its second half. This paper has not been peer reviewed.

2 Two nulls, and the fact that they nest

Definition 2.1 (Four-point condition)**.** A symmetric d satisfies the *four-point condition* when for all a, b, c, e, not necessarily distinct,

$$d(a, b) + d(c, e) \;\le\; \max\big(d(a, c) + d(b, e),\; d(a, e) + d(b, c)\big). \tag{2}$$

A quadruple admits three pairings; (2) says no pairing's distance sum strictly exceeds both others. This is a statement about tree structure with no constraint on rates: branch lengths are free.

Theorem 2.2 (Tree-Metric Theorem; Buneman. *Cited, not proved here*)**.** *Let δ be a nonnegative dissimilarity map on a finite set — zero on the diagonal, symmetric, non-negative. Then δ is a tree metric if and only if it satisfies* (2) *[1, 2].*

The hypotheses are weaker than "metric", and the distinction matters enough to state: setting $y = z$ in (2) yields the triangle inequality, so the triangle inequality is a *consequence* of the condition and requiring it as a hypothesis would state something strictly weaker than the theorem. An earlier draft of our own documentation made that error.

Theorem 2.3 (The nulls nest)**.** *Every symmetric ultrametric satisfies the four-point condition.*

Proof. Fix a, b, c, e. Four applications of (1), with symmetry to order each application, give $d(a,b) \le \max(d(a,c), d(b,c))$, $d(a,b) \le \max(d(a,e), d(b,e))$, $d(c,e) \le \max(d(a,c), d(a,e))$ and $d(c,e) \le \max(d(b,c), d(b,e))$. Each is a disjunction, giving 2^4 cases; splitting further on the order of the four bounding quantities, every branch bounds $d(a,b) + d(c,e)$ by one of the two right-hand sums of (2) by a linear arithmetic step. The case analysis is mechanical and is discharged by the kernel (Listing 1). □

Corollary 2.4 (What a quartet violation buys)**.** *If d is symmetric and fails* (2) *then d is not ultrametric either.*

Corollary 2.4 is the licence for the reading that organises Section 3: large triple margins together with floor-level quartet margins indicate rate variation on a tree, because the nesting leaves that the only region available.

Listing 1: `formal/ANSE/GeneticaReticulataTree.lean`. The `rcases` chain is the 2^4 case split, with four further splits on the order of the bounding quantities.

```
def IsFourPoint (d : α → α → ℚ) : Prop :=
  ∀ a b c e, d a b + d c e ≤ max (d a c + d b e) (d a e + d b c)

theorem fourPoint_of_ultrametric {d : α → α → ℚ}
    (hsym : ∀ x y, d x y = d y x) (hu : IsUltrametric d) : IsFourPoint d
```

The module carries 18 declarations through an axiom gate that compiles each and runs `#print axioms`: zero `sorry`, and no axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}. Checking compilation alone is insufficient — both `sorry` and a locally declared axiom exit zero — which is why the gate inspects the axiom list.

3 The clock-level statistic is refuted by a control with no reticulation

Condition (1) needs three sequences rather than four, so it is cheaper, and our first version reported it. It separated a simulated clonal population from a simulated recombinant cleanly. The separation was real and the inference from it was wrong, for the reason Modelling assumption 1.1 warns of: a violation tells you one of two assumptions failed, not which.

The decisive control is a **divergence-matched relaxed clock**: clonal descent on a balanced tree, no reticulation whatever, with each lineage's substitution rate multiplied by e^{su} for u uniform on $[-1, 1]$. The matching is not optional. Since $\mathbb{E}[e^{su}] = \sinh(s)/s > 1$, an unnormalised relaxed clock is also a *faster* clock, and a violation could then be attributed to saturation rather than to rate variation; dividing by that mean leaves expected divergence equal to the strict-clock case while realised rates vary widely. In our first run the normalisation changed the answer, taking the relaxed-clock quartet violation count from 2/5 seeds to 0/5.

Read rows two and three of Table 1 together. The smallest triple margin produced by a population with *no reticulation in it at all* is 99/4000; the largest produced by a population

Table 1: Thirty-two leaves, 4000 nt, three seeds, exhaustive over all 35 960 quadruples per dataset. Margins are exact rationals with denominator 4000; numerators given seed by seed.

simulated population	reticulation?	clock	triple margin	quartet margin
clonal	no	strict	$24, 30, 29$	$10, 9, 10$
clonal	**no**	**relaxed**	$\mathbf{155}, \mathbf{157}, \mathbf{99}$	$11, 18, 8$
one recombinant	yes	strict	$79, 77, 75$	$79, 70, 69$
two reciprocal mosaics	yes	strict	$79, 77, 75$	$\mathbf{158}, \mathbf{154}, \mathbf{126}$

containing a real recombinant is 79/4000. The ordering is inverted at every seed. On this data a larger triple margin is weak evidence *against* reticulation. The quartet column does not move: the relaxed clock stays at the strict clock's level while the reciprocal-mosaic positive control separates from it by a factor of 12.6 at its weakest seed, with no overlap. At sixteen leaves the same experiment gives a relaxed-clock triple margin reaching 106/4000 — above every single-recombinant margin — exceeding twice the clonal floor at 3 of 5 seeds, while the quartet statistic does so at 0 of 5.

Adding taxa makes the triple statistic worse, as the mechanism predicts: it is a maximum over triples, so more leaves give more chances at an extreme pair of rates.

Remark 3.1. This confound is not peculiar to our statistic. Frankel and Ané report that rate variation across lineages produces a marked increase in the false discovery of reticulation for three summary tests of introgression, with the D_3 statistic reaching about 80% type-I error, and read this as D_3 responding more strongly to a departure from the clock than to reticulation itself [8]. A null that assumes a clock is vulnerable wherever it is used. The established test whose null assumes no clock, Bruen, Philippe and Bryant's Φ_w, reports 5.7% false positives on a non-recombinant control where NSS reports 37.5% [6].

Recommendation. Report the quartet statistic as primary. Keep the triple statistic — it is cheaper, and a triple violation with *no* quartet violation is itself informative, being the signature of rate variation on a tree — but a triple certificate alone is not evidence of reticulation.

4 Missing data: complete deletion within the subset

Pairwise deletion computes each pair's distance over the columns comparable in that pair, giving different pairs different denominators, and that alone can break (1) on a genealogy that is exactly a clock tree. The fix is *complete deletion within the subset*: count only columns at which every member of the triple or quadruple carries a definite base. All distances in a certificate then share one denominator, so the obligations become comparisons between integers with no cross-multiplication — a smaller trust surface and the repair at once. The cost is power: at a 25% gap load roughly 1000 to 1700 of 4000 columns survive within a triple.

Measured on gap-blocked but strictly clonal, strictly clock-like populations: pairwise deletion calls 33 to 52 of the 560 triples violations that complete deletion within the triple denies, at every one of five seeds; at thirty-two leaves, 286 to 366 of 4960. With no missing data the two policies agree on all 560, as they must.

Remark 4.1 (Margins under the two policies are not comparable). Complete deletion shrinks the denominator, so one differing site is worth more and its margins come out *larger* although the policy is strictly more conservative. A criterion comparing the two sets of margins was written into our first run and withdrawn. Compare per-subset *verdicts*, not magnitudes.

Table 2: The admissibility ladder. Sixteen taxa, 4000 nt, five seeds, 1820 quadruples per seed (9100 per rung). Every alignment certified optimal (Section 6).

mean deletion length (nt)	inferred gap fraction	worst column recovery	quadruples manufactured	seeds with any
30	0.00000	1.00000	0	0/5
30	0.00398	0.99997	0	0/5
30	0.00549	0.99880	251	2/5
30	0.00870	0.99712	108	1/5
30	0.01507	0.99906	110	3/5
30	0.02012	0.99831	149	3/5
120	0.00000	1.00000	0	0/5
120	0.01322	0.99901	57	2/5
120	0.01627	0.99901	54	2/5
120	0.03731	0.99467	324	3/5
120	0.05254	0.97501	253	3/5
120	0.08233	0.96032	**1600**	5/5

5 The alignment step is on the trust surface

A certificate is machine-checked, which invites the claim that the conclusion does not depend on the pipeline. The precise statement is narrower. The kernel is handed the sequences and recomputes every count from them, so a false inequality, a wrong count, a transposed pair or a miscomputed margin is excluded. What remains trusted is that the embedded rows are the data they are claimed to be — and on real data those rows come out of an **aligner**. If the alignment places non-homologous residues in the same column, the counts are computed faithfully from the wrong data and the kernel accepts them. Proof moves the trust boundary; it does not close it.

That cannot be proved away, so we measure it. We simulate clonal, relaxed-clock descent with deletion events in a fixed coordinate frame, so the frame *is* the true alignment and **there is no reticulation in the population at all**; strip the gaps; hand the ragged sequences to the aligner; realign; and compare per-quadruple *verdicts*, not margins, since the two alignments have different widths. Any quadruple the inferred alignment calls a violation and the true alignment does not was manufactured by the aligner.

The aligner is reference-anchored: each sequence is aligned to a centroid reference by affine-gap global dynamic programming and projected onto the reference's coordinates. This is the construction viral genomic surveillance uses, and the final step of MUSCLE5's `Super5` mode. Its one structural advantage is that without a guide tree there is no error-propagation failure mode; its costs are that insertions relative to the reference are dropped and that every pair of non-reference sequences is related through the reference.

At the heaviest rung the aligner invents 1600 of the 9100 quadruple verdicts on a genealogy that is exactly a tree. **Past a threshold, alignment error is not a correction to this test; it is the result.** The curve is not monotone, which is what a maximum over 1820 quadruples at five seeds looks like, and is one reason to gate on *whether* anything was manufactured rather than on how much.

So the admissibility threshold is read off the ladder rather than chosen: a gap fraction of 0.00398 and 8.06×10^{-4} dropped insertions per aligned cell, the heaviest rung at which nothing was manufactured at any seed. Both are *rates*; an absolute insertion count is not comparable between a 3 kb and a 12 kb genome.

Why we believe this measurement is new. That misalignment manufactures recombination signal is not a novel claim and not a contested one. It is stated outright in the RDP5 manual — *"misalignment of sequences is a major cause of false recombination signals"* and *"all recombination signal detection methods are extremely sensitive to misalignment and will usually identify misaligned regions of sequence as having a recombinant origin"* — and misalignment artefact is a named output category in that program [10]. What we did not find is the quantification. Across the recombination benchmarking literature we examined, every study varies divergence, rate heterogeneity, topology, selection or sequence count while holding the alignment fixed; the most recent full benchmark we reached does not discuss gaps at all [11]. A measurement of the form "alignment error alone manufactures N of M apparent signals on data containing no reticulation" appears therefore to be the first direct measurement of a mechanism the field already accepts. That is confirmatory-but-novel, and we state it that way rather than as a discovery. The nearest quantified precedent we found is in an adjacent task: aligner choice driving a reported 48–82% of false positives in *Drosophila* positive-selection scans.

An honest limitation of the simulation. The deletion rate and mean deletion length are *chosen*, not calibrated: we found no published viral indel length distribution and no clean per-site viral indel rate. The ladder therefore varies a load; it does not match one to biology.

6 Certifying the band: a proxy inside our own apparatus

The dynamic programme is banded, because 12 kb against 12 kb is 1.4×10^8 cells and the optimal path of two homologues does not wander far from the diagonal. A band is a cap on how much indel a pair may be assigned, so it is a place where an aligner can silently return a sub-optimal path. Our first implementation detected when the optimal traceback *touched* a band edge, widened the band until it did not, and documented this as a guarantee that no constrained path was ever returned without saying so.

That documentation was wrong. Gibrat measures both failure directions on real data: 2.2% of alignments are optimal yet reach the band edge, and 0.5% are sub-optimal without ever reaching it [4]. Not touching the edge is evidence of optimality, not proof of it.

The repair needs no theorem. Computing the unbanded optimum by full Gotoh dynamic programming in $O(nm)$ time and $O(m)$ memory — no band, no heuristic, no traceback — and refusing any banded path that does not attain it gives a certificate for one extra pass per pair, which at fifteen pairs per population is affordable. We also note, and do not use, a cheaper sufficient test: Gibrat gives an upper bound on the score of any path leaving the band, so attaining that bound certifies optimality in constant time. His derivation is for a band about the main diagonal while ours is about the proportional diagonal, so the bound does not transfer unexamined, and recomputation needs no derivation to be trusted.

The proxy had really failed. The first certified run exited non-zero immediately, on a pair scoring 16555 against an unbanded optimum of 17275 — a deficit of 720 points, with the traceback never touching an edge, so edge-driven widening never fired. The widening loop is now driven by the *score*: widen on either signal, and return only a path that attains the optimum, or an error.

What it had inflated. With every alignment proved optimal, Table 2's heaviest rung manufactures 1600 rather than 4876, and worst column recovery is 0.960 rather than 0.868. Roughly two thirds of what we had reported as reference-anchored alignment error was our own unchecked heuristic. The qualitative conclusion is unchanged and the thresholds do not move, so no verdict in Section 12 changes; but the magnitude was overstated threefold, in the direction that makes a finding look more impressive. We record this because it generalises: **an**

uncertified step inside a measuring apparatus inflates the thing being measured, and it will tend to inflate it favourably, which is the direction least likely to prompt a second look.

Remark 6.1 (A textbook algorithm that is wrong in print). While checking the above we found that Gotoh's *published* initialisation is incorrect — it omits one of two gap-opens and can return a score that is not the cost of any alignment — and that the error is reproduced in standard textbooks and in five of ten inspected implementations [5]. Ours turns out to be the corrected form, but nothing in our codebase established that until the dynamic programme was tested against exhaustive enumeration of every alignment of short sequences rather than against another dynamic programme.

Remark 6.2 (A tie-break that is a free parameter). The control that the aligner is the identity on gapless input failed at one seed of five on our first implementation, and not because of a bug. With an affine gap of -10 to open and -1 per residue, three consecutive mismatches score $3 \times (-4) + 5 = -7$, and deleting one reference residue, aligning the rest, then inserting one query residue scores $-11 + 15 - 11 = -7$. The identity was one of two optimal alignments and the implementation returned the other. The repair was to *state* the preference. An aligner has free parameters that no amount of downstream proof constrains.

7 A maximum, where the established tests are averages

Our certificate is a maximum over quadruples: the single most-violating quadruple, exhibited as a witness the kernel can check. That is what makes a certificate possible, and it is also an extreme-value statistic whose null level grows with the number of subsets examined. Our own grid shows it — the quartet floor moves from 5–7 at sixteen taxa to 9–10 at thirty-two — so margins are not comparable across populations of different size.

The treelikeness literature took the other branch: its statistics are averages over quadruples, among them $\delta_q = (m_1 - m_2)/(m_1 - m_3)$ for sorted pairing sums $m_1 \geq m_2 \geq m_3$ [7]. We therefore ran the comparison.

Table 3: Seed-paired against the hard null (clonal, relaxed clock, no reticulation). Separation is the weakest seed's ratio of the reciprocal-mosaic control to that null.

statistic	what it is	beats the null	worst-seed separation
quartet max margin	exact, with a witness	5/5	**16.57**×
mean δ_q	the averaged statistic	5/5	5.07×
proportion violating	share of quadruples	5/5	1.22×

All three rank both reticulate populations above the relaxed-clock null at every seed, so none of them inherits the confound that destroyed the triple statistic. At fixed taxon count the maximum separates more than three times as widely as the mean, and the proportion-based variant barely separates at all. We note that this reproduces, independently, a published finding that the proportion-of-resolved-quadruples variant fails as a test while the averaged variants hold a correct false-positive rate.

What the comparison does *not* settle is the maximum's actual weakness, which is incomparability across taxon counts rather than poor separation within one; our grid is all at one size. Both statistics are therefore reported, and only the maximum is exact and kernel-checkable, because a mean has no witness to exhibit. That is the trade, and it is now priced rather than assumed.

8 The head-to-head against the incumbent

A method should be compared with what it hopes to improve on, and edition 1 of this paper did not do that. The relevant incumbent is not one of the many tests that assume a clock — those fail the control of Section 3 by construction — but the one whose null assumes **site-order exchangeability** and nothing about rates: the pairwise homoplasy index of Bruen, Philippe and Bryant [6]. Like the four-point condition it is immune by design to the relaxed-clock confound, so a comparison between them is about power and about what each licenses, not about survival.

We implement a restricted form. Two biallelic sites are *incompatible* when all four combinations of their states occur among the sequences — the four-gamete test. Under a single tree, compatibility between two sites does not depend on their separation in the alignment; under recombination it does, since neighbouring sites tend to share a local history. The statistic Φ_w is the mean compatibility of informative sites within a window w, and the null comes from permuting the site order. Two restrictions are stated rather than hidden: only biallelic parsimony-informative columns are used, so the four-gamete test is exact and needs no multistate refinement; and columns with a gap in any member are skipped, which is the same complete-deletion policy the certificate uses, so both tests see the same data. This is not the published statistic in full and we do not present it as such.

One property of the implementation is worth noting. Φ_w is a count of compatible neighbouring pairs over a count of pairs, and the pair count is invariant under permutation of the site order, so the permutation test compares *numerators*: it is exact integer arithmetic, like the certificate, rather than a normal approximation.

Table 4: All three detectors on the same alignments, sixteen taxa, five seeds, counts of seeds significant at 0.05. Φ_w: window 50 informative sites, 199 permutations, floor 1/200. The quartet column is the certificate's margin against a floor *matched* to each population's own width, divergence and gap load, decided by exact integer cross-product. Mean δ_q is calibrated by parametric bootstrap under the same matched null, 99 replicates, floor 1/100.

population	retic.	Φ_w	quartet vs matched	mean δ_q
clonal, strict clock	no	0/5	**0/5**	0/5
clonal, **relaxed clock**	no	**0/5**	**0/5**	0/5
one recombinant	yes	5/5	**5/5**	5/5
relaxed clock + one recombinant	yes	5/5	**5/5**	5/5
reciprocal mosaics	yes	5/5	**5/5**	5/5
gap blocks	no	**no measurement**	**0/5**	0/5
saturated	no	**0/5**	**0/5**	0/5

The result, stated plainly. All three detectors agree on all seven populations — specific on the four without reticulation, detecting all three with it at every seed — except that Φ_w cannot be measured on the gap-blocked data at all. **The incumbent is not beaten on power, and this method's case cannot rest on power.** Edition 1 framed the missing comparison as the validation we lacked; it is a three-way tie, and that is the more useful finding.

Edition 2 of this table was wrong in our own favour's opposite direction, and the correction is worth stating. It judged the certificate against the *strict-clock* population's floor while judging the competitors against matched nulls, and so recorded the certificate as firing 5/5 on the gap-blocked and saturated populations — neither of which contains reticulation. That was the incommensurability this paper warns about in Section 4, committed a fourth time, against ourselves: the gap-blocked population has a different denominator and the saturated one ten

times the divergence. Given a floor matched to its own width, divergence and gap load, the certificate is specific on all four. Nothing about the method changed; the comparison was corrected. The third column is the calibrated averaged statistic (parametric bootstrap under a matched tree null), included because it is the fairest possible competitor: same quadruples, same alignment, same matched-null discipline, no permutation.

Two asymmetries, one each way. The incumbent needs no recalibration when divergence changes: it is specific on the saturated population at ten times the divergence, where the certificate's raw margin against the strict-clock floor fires at 5/5. We are careful about what that 5/5 is. The saturated population's divergence and denominator are not the floor's, so the comparison is not commensurable, and this is the same error — in a fourth guise — as comparing margins across different denominators; the matched-floor construction of Section 12 exists precisely for it. But *needing* a matched null is a cost the incumbent does not pay, and that is a fair point against the certificate in deployment. Conversely, on the gap-blocked population the incumbent returns **no measurement at all**, having too few biallelic gap-free informative columns once complete deletion is applied, where the certificate still works.

What is left once power is off the table. Two things, and we think they are worth more than a power claim. First, a certificate is an **exhibitable witness**: four sequences and six counts that a kernel recomputes, so a reader re-verifies the finding without running our search or trusting our description of it. Second, and specific to this paper's own measurement: the null is a theorem rather than a permutation. The RDP5 manual states that misalignment-induced false positives are detected with significant p-values in the real alignment yet *"will likely be undetectable in the shuffled alignments"* [10]. A site-order permutation null — which is exactly this incumbent's null — is therefore blind by construction to the artefact class Section 5 measures. Neither point is established by Table 4; both are the actual case for the approach, and stating them in place of a power claim is the honest position.

9 Published breakpoints, and the resolution limit

Section 12's populations rest on two weak things: a literature *expectation* that a family recombines, and an alignment from the aligner of Section 5, whose error had to be measured and gated on. Both can be replaced, and doing so locates this method's resolution limit.

The Pango designation process records SARS-CoV-2 recombinants as `X*` lineages with their parent lineages and, for many, an explicit numeric breakpoint interval in reference coordinates, each traceable to a reviewed proposal; and Nextstrain publishes an open release *already aligned* by Nextclade to MN908947, the frame those coordinates are quoted in. Two triads were built, 16 genomes each (five from each parent lineage, six from the recombinant), under filters fixed in advance.

Table 5: Published-breakpoint populations. The prediction, registered before the test, is that the certificate's dominant pairing separates the two *parent* lineages with a recombinant on the other side — a constraint on *which* four sequences are picked, which a margin cannot supply.

recombinant	parents	breakpoint	prediction	margin
XCP	EG.5.1.10 × JD.1.1	14190–14633	**held**, 6/10	2/28920
XCM	XBB.2.3 × DV.7.1	22017–22031	**failed**, 0/10	4/28511

On XCP the method picked the two true parents as the far pair in six of its ten strongest certificates. On XCM every certificate instead paired a parent with a recombinant on each side.

One of two is a split result and is reported as one.

The finding is a resolution limit. Both margins are at the edge of what the statistic resolves. XCP's population has mean p-distance 0.00057 and XCM's 0.00127, implying roughly 17 and 38 differing sites across 29 903 columns; the margins are 2 and 4 *sites*. A verdict cannot rest on that however the floor comparison comes out — and XCP's floor comparison is itself degenerate, the matched null having produced no violating quadruple in any replicate, so "exceeds the floor" reduces to "the margin is positive". Both the degeneracy and the site counts are now computed into the output rather than left to prose.

So, stated against our own interest: **SARS-CoV-2 recombinants between recently-diverged lineages sit below this method's resolution.** And there is a tension worth naming because it will recur — **the datasets with the best ground truth are the least divergent**, since recombinants are named between lineages close enough to be confidently assigned. Section 12's populations, at mean p-distance 0.0088 and 0.11, suit the statistic far better and are far worse supplied with truth.

One signal survives, labelled as suggestive rather than evidence: on XCP the *pairing* was right in 6 of 10 certificates, in a regime where the magnitude means nothing. The method pointed at the correct two parents.

A note on the trust surface. Using a published alignment removes *our* aligner and substitutes Nextclade's, which is reference-anchored too — the same construction and so the same class of error Section 5 measures. Provenance improves; measurement coverage does not, since we have not measured their error. The admissibility gate calibrated for ours does not transfer and is not applied.

10 The maximum statistic's defect, measured across taxon counts

Section 7 concluded that the certificate's weakness is not power but incomparability across taxon counts, and argued it from two data points. This paper has twice recorded the cost of reasoning from too few points, so the defect is measured.

That required making the measurement affordable. Sixty-four taxa is 635 376 quadruples and the byte-at-a-time counting visits 4000 columns per quadruple. Packing each sequence into three bitvectors over columns — the base's two bits, plus a mask of definite bases — turns a quadruple's seven counts into one pass over $L/64$ words: intersect four masks for the common columns, then count differing bits for each of the six pairs inside it. Measured 21.6× faster at thirty-two taxa with identical exact margins. It is a second implementation, not a replacement: the byte path remains the specification, agreement is a property test across widths 1, 7, 63, 64, 65, 130 and 257 plus gap-blocked data, and the sweep re-checks it on every run.

Table 6: Exhaustive over every quadruple, three seeds, margins over 4000. The "above floor" column is seed-paired and decided by exact integer cross-product.

taxa	quadruples	clonal floor	one recombinant	above floor	mosaics
16	1 820	7, 5, 6	60, 52, 53	3/3	136, 112, 124
32	35 960	10, 9, 10	60, 67, 98	3/3	148, 154, 108
64	635 376	17, 16, 14	89, 82, 83	3/3	156, 148, 166

The floor roughly doubles from sixteen to sixty-four taxa while the signal stays flat, so the worst-seed separation of mosaics over the floor halves: 19.4×, 10.8×, 9.2×. Two readings, both

needed. Detection at a *fixed* taxon count survives comfortably. But "twice the floor" means something different at sixteen taxa than at sixty-four, so a threshold calibrated at one size does not transfer — which is why Section 12's verdicts use a floor matched to each population, and why edition 2's raw-margin comparison was the wrong one.

One figure looks alarming and is not: at sixty-four taxa 82% of all quadruples violate the four-point condition on a clonal, strict-clock population. Finite sequences are never exactly additive, which is precisely why the *margin* and not the *count* carries the inference — and why the proportion-violating variant of Section 7 separated by only 1.22×.

11 A topology experiment, and the grounds for closing it

The construction this programme began from reads an unfilled 1-cycle in the Vietoris–Rips complex of a genetic-distance cloud as reticulation, on the same warrant as our null: a tree metric admits no unfilled cycle. A search of the literature found no peer-reviewed independent calibration of that method's specificity. Our grid has the ground truth to supply one, since the saturated dataset differs from the baseline only in having ten times the substitution rate.

At sixteen taxa the experiment is **inconclusive** and its output says so (`"this_experiment_is_interpretable": false`): the matched null produced no cycle in 495 of 495 replicates on five of seven datasets, so the p-value takes only two values and the test reduces to *did a cycle appear at all*. The cause is diagnosable — at that size the complex is too small for H_1 to populate — and the fix is more taxa, which §10 made affordable.

Re-run at sixty-four taxa, the answer arrives. Two nulls become non-degenerate, and they are the two datasets containing *no reticulation*: the statistic fires at 3/3 on gap-blocked data and at 3/3 on saturated data, while a genuine single recombinant produces **no cycles at all** and is missed at 0/3.

Five hypotheses were then written down for that failure, three selected on cost-to-decide, and the criterion fixed in advance: a variant succeeds if the reciprocal-mosaic population is significant at every seed and both no-reticulation populations at none.

Table 7: Filtration funnel, sixty-four taxa, three seeds, 49 bootstrap replicates. No variant meets the criterion.

variant	hypothesis	mosaics	one recomb.	gaps	saturated
baseline, total	—	3/3	**0/3**	2/3	3/3
F1 Jukes–Cantor	saturation	3/3	**0/3**	2/3	3/3
F2 global complete deletion	denominators	3/3	**0/3**	**0/3**	3/3
F3 longest bar	the summary	3/3	**0/3**	**0/3**	3/3

F1 is refuted: correcting for saturation leaves that cell at 3/3 and *raises* its statistic, 0.00867 → 0.01421. **F2 is confirmed**, and confirms the construction argument below: global complete deletion takes gap-blocked data from 13.3 cycles and 2/3 to 0.0 cycles and 0/3, so the gap false positive was the missing-data confound of Section 4 reappearing as topology. F3 removes it too, by distinguishing many weak cycles from few strong ones. But **the miss is not a filtration artefact** — zero cycles on a single recombinant in all four variants — and the certificate detects that same population at 3/3 with roughly fivefold margins (Table 6).

Decision: the line is closed. Not because topology is the wrong idea; the warrant, *a tree metric admits no unfilled cycle*, is the same fact this paper's null rests on. What is closed is the global-distance-matrix implementation, for two reasons the funnel established rather than assumed. It **cannot use per-subset complete deletion**, because a complex is built from all pairs at once and there is no "within this subset" to restrict to; F2 shows the available substitute

works but costs every column any taxon is missing. And it **misses the signal the certificate catches**. So this programme's topological content is the certificate itself: the same theorem, evaluated combinatorially per quadruple, which is exactly what makes complete deletion within the subset available at all.

Limits. Three seeds, one generative model, and the saturated cell's null is degenerate in every variant, so that column carries no calibration and its 3/3 is the binary above. The two conclusions the decision rests on — F1's refutation and the single-recombinant miss — do not depend on it.

Remark 11.1 (The lesson both of these share). Two experiments in this edition produced a confident-looking count from a null that could not vary. A null that is identically zero makes every positive observation significant at the floor, which is indistinguishable from a powerful test until you look at the null. Both now compute a degeneracy flag, and no calibrated statistic in this work should have its p-value read before that flag is checked.

12 Real populations

Five viral taxa were drawn from a public corpus of 58 046 viral genomes with taxonomy [12] under filters fixed in advance: within 2% of the taxon's median length, at most 0.1% ambiguous residues, exact duplicates removed, sixteen sequences per population, selected either in accession order or by greedy 6-mer diversity. Each taxon carries a *literature* expectation about whether its genealogy is tree-like, written before any margin was computed, and the test is free to contradict it. Ten populations result.

Seven fail the admissibility gate of Section 5 and are reported blocked with their counts rather than interpreted. This is expensive on purpose: one discarded population has a margin of 30/1784 and another 13/5082. Note also which selection strategy survives — diversity maximisation is exactly what breaks reference-anchored alignment. Each of the three that pass is aligned twice, on the two most central sequences, and each margin is compared against a floor from a simulated clonal relaxed-clock tree at *matched* alignment width, *matched* mean p-distance (tuned by bisection) and *matched* gap load, because a margin alone is not a statistic and a floor from the wrong divergence measures the floor.

Table 8: The three admissible populations. References agree on the verdict in all three.

population	expectation	margin	matched floor	verdict
Hepatitis B, accession order	reticulate	9/3182	2/3182	above 2×
Coxsackievirus A6, diversity	reticulate	329/7312	268/7390	above 1× only
JC polyomavirus, accession order	tree-like	4/5108	24/5121	**below**

Three populations, three agreements with the prior. The JC polyomavirus row is the one that could most easily have embarrassed the method and did not: its margin is a sixth of the floor a tree produces at matched width and divergence. Both populations clearing their floor carry a Lean file embedding the four aligned rows, so the kernel recomputes the six pair counts and the common column count itself: accepted at 3182 columns in 52.7 s and at 7434 columns in 117.6 s, with both corrupted twins rejected.

Two cautions belong beside Table 8 rather than in a footnote. Three is not a validation. And the Coxsackievirus row is weak and at high divergence (mean $p = 0.11$): it clears 1× and not 2×, and the noise floor itself grows with divergence, so it should be read as not inconsistent with the prior rather than as a detection. A further caution concerns the prior itself: one of our two negative-control families, the papillomaviruses, has documented within-type recombination [9],

so that prior is the weakest in the set and we mark it as such. A clonal bacterial lineage would be the stronger negative control.

13 The Gromov hinge of Buneman's converse

Theorem 2.2 is cited above and not proved. The constructive proof in the literature does not attack the tree directly: it factors through the *Gromov product* at a chosen root, reducing the four-point condition on d to an ultrametric-style condition on a derived function, and only then builds an equidistant tree by a cherry-picking recursion. The reduction is the hinge, and unlike the tree construction it is pure algebra.

Definition 13.1. At a root r, write $g_r(x,y) = \frac{1}{2}(d(x,y) - d(r,x) - d(r,y))$. This is a dissimilarity, not a metric: it takes negative values. Say f satisfies the *strong triangle inequality* when $f(x,z) \le \max(f(x,y), f(y,z))$ for all x, y, z — inequality (1) with no positivity required.

Theorem 13.2 (The hinge; proved, not cited)**.** *A symmetric d satisfies the four-point condition if and only if g_r satisfies the strong triangle inequality for every root r.*

Proof. Expanding $g_r(x,z) \le g_r(x,y)$ gives $d(x,z) + d(r,y) \le d(x,y) + d(r,z)$, and $g_r(x,z) \le g_r(y,z)$ gives $d(x,z) + d(r,y) \le d(y,z) + d(r,x)$. So the right-hand side of the theorem says exactly

$$d(x,z) + d(r,y) \;\le\; \max\big(d(x,y) + d(r,z),\; d(y,z) + d(r,x)\big)$$

for all r, x, y, z, which is (2) instantiated at $(a,b,c,e) = (x,z,r,y)$ together with the symmetry of d. Reading the implication backwards requires the root c with the remaining three in the order (a,e,b). □

Listing 2: The Lean statement. `#print axioms` gives `[propext, Classical.choice, Quot.sound]`.

```
theorem fourPoint_iff_gromov_strongTriangle {d : α → α → ℚ}
    (hsym : ∀ x y, d x y = d y x) :
    IsFourPoint d ↔ ∀ r, StrongTriangle (gromov d r)
```

Two details are worth recording. The condition is named `StrongTriangle` and not `IsUltrametric`, because the Gromov product takes negative values and calling it an ultrametric would import a positivity hypothesis it does not satisfy; the literature's phrase "the Gromov product is an ultrametric" means only the max inequality. And the kernel earned its place: our first attempt instantiated the reverse direction at the wrong permutation, `linarith` failed on two goals, and the axiom gate reported `sorryAx` rather than letting a plausible-looking proof through.

What remains cited is the tree construction — the equidistant cherry-picking recursion and the re-rooting that returns a tree metric. The chain this paper can claim is therefore: certificate ⇒ not four-point ⇒ (proved) some root's Gromov product fails the strong triangle inequality ⇒ (cited) not realised by a weighted tree. We searched Mathlib, the Isabelle Archive of Formal Proofs and the Coq ecosystem and found ultrametric spaces but no tree metrics, no splits and no four-point condition, so completing the construction appears to be first of its kind. That search was across indexes rather than exhaustive.

14 The geometry the certificate was already doing

Section 11 closed the Vietoris–Rips route and concluded that this programme's topological content is the certificate itself. That is a slogan unless it is made precise. It is precise, and it is a theorem.

Classical phylogenetic combinatorics assigns each split S of a quadruple a **Bandelt–Dress isolation index**

$$\alpha(S) \;=\; \tfrac{1}{2}\big(\min\{\text{the other two pairing sums}\} - S\text{'s own pairing sum}\big),$$

in distance units. A split with $\alpha > 0$ is one the data supports, and on a tree metric the supported split's index is the internal edge's length. This is the basis of split decomposition and of split networks — the canonical geometry of tree-likeness, and the object a distance filtration was a poor substitute for.

Theorem 14.1 (The margin is an isolation-index gap)**.** *Let $s_1 \le s_2$ and $s_1 \le s_3$ be a quadruple's three pairing sums, s_1 minimal. Then*

$$\alpha(s_2) - \alpha(s_3) \;=\; \tfrac{1}{2}(s_3 - s_2),$$

and the four-point condition on that quadruple holds exactly when this gap is non-positive.

Proof. Both non-minimal splits have s_1 as the minimum of their other two sums, so $\alpha(s_2) = \frac{1}{2}(s_1 - s_2)$ and $\alpha(s_3) = \frac{1}{2}(s_1 - s_3)$; the difference is $\frac{1}{2}(s_3 - s_2)$. The four-point condition reads $s_3 \le \max(s_1, s_2) = s_2$, which is exactly $s_3 - s_2 \le 0$. □

Since the certificate's margin is $s_3 - s_2$ over the common column count, Theorem 14.1 says the margin is **twice an isolation-index gap**, and the condition the certificate refutes is the vanishing of that gap. The number the method reports is a split-decomposition quantity, not merely analogous to one. Both statements are machine-checked (`isolation_gap_eq_half_sum_gap`, `quadruple_fourPoint_iff_isolation_gap_nonpos`).

Note what the proof does not use: no tree, no complex, no metric space, only three rationals. That is exactly why the certificate can be evaluated per subset where a complex over a global distance matrix cannot, and so why Section 11 closed an implementation rather than an idea.

Checked against the data, not only proved. The theorem is about three rationals. The implementation computes each split's index from its own definition — the minimum of the *other two* sums minus its own — rather than from the sorted sums, so a mistake would surface as a mismatch rather than cancel out, and the identity is then checked on every quadruple examined. Across all seven datasets at five seeds it holds on 75 500 quadruples and fails on none.

And the measurement that follows. Reporting the split decomposition's own headline quantity is informative in a way worth recording. The largest supported-split index is *identical to five decimal places* on the clonal tree and on both reticulate populations built from it — 0.02863, 0.02637, 0.02738 across the first three seeds for the clonal population, the single recombinant and the reciprocal mosaics alike. It is blind to the reticulation, which it should be: the strongest supported split is a deep split of the underlying tree, and adding recombinants does not disturb it. What it tracks instead is divergence — the saturated population's index is seven times the others'.

So the statement this paper arrives at is sharper than "the certificate is topological":

> The split decomposition's supported index carries the *tree*. The gap between the non-supported indices carries the *reticulation*. The certificate measures the gap.

That also explains, after the fact, Section 11's failure. A Rips filtration is driven by distance *magnitudes*, which is the quantity that tracks divergence; the signal lives in a *difference* between indices that magnitudes do not expose. A statistic that fires on saturated data and misses a single recombinant is what that predicts.

15 Limitations, and what would make this wrong

- **Buneman's converse is half-cited.** Theorem 13.2 closes the first step; the construction is not formalised. Likewise Theorem 14.1 identifies the margin with an isolation-index gap but does not prove that a supported split's index is a tree edge's length, which needs the same construction. So "satisfies (2) everywhere" is not yet formally "is a tree" in this development.
- **A certificate does not identify an event.** Satisfying (2) does not certify the *absence* of reticulation, and a violation does not pin down which tree the data departs from; there exist networks whose distances are representable by a tree other than their own underlying tree [3]. The honest report is *that* the data are not tree-like and *where*, not an event history.
- **The margin's null distribution is not derived.** Finite sequences are never exactly ultrametric, so certificates exist under the null with small margins; we calibrate by matched mechanistic simulation rather than by a distribution. We found no analytic sampling distribution of the four-point margin in the literature either, only high-probability sequence-length bounds and parametric bootstrap, so this is the field's position and not only ours.
- **Three admissible real populations.** Too few to call the method validated, and the aligner that made them admissible is reference-anchored.
- **The head-to-head is a tie, not a win** (Section 8), and it covers one incumbent. The comparison against the quartet-averaged treelikeness statistics *used as calibrated hypothesis tests* rather than as scores has still not been run, nor against the clock-assuming tests, which would fail Section 3's control but are what most practitioners actually use.
- **The statistic is an extreme value.** Section 7 prices this at fixed taxon count only.

What would make the programme wrong, rather than merely incomplete: if violations of (2) at the scale we report turned out to be dominated by some departure from tree-likeness other than reticulation — population structure, recurrent mutation, or selection acting locally — then the certificates would remain valid and the biological reading of them would not. We have controlled for rate variation and for missing data, and measured the alignment contribution. We have not controlled for the rest.

16 Reproducibility

Every number in this paper is produced by a named command in the repository `SocrateAI-Scientific-Genetic` at release `v1.8.0`, and the raw output is deposited with it.

command	what it produces
`reticulation experiment`	Table 1, Section 4, Table 3
`reticulation align-error`	Table 2 and the admissibility threshold it measures
`reticulation real-data`	Table 8 and the kernel verdicts
`reticulation saturation`	the length-scaling diagnostic behind the divergence caution
`reticulation experiment --phi-window`, `--phi-permutations`	Table 4; defaults 50 and 199
`reticulation scale`	Table 6 and the 21.6× check
`reticulation filtration`	Table 7
`reticulation breakpoint`	Table 5
`scripts/reticulata_hf_populations.py`	the ten populations, with a SHA-256 of the source file
`lake build ANSE.GeneticaReticulataTree`	the theorems, with `#print axioms` on each

All are deterministic given their seeds. The population extraction is deterministic given the source corpus, whose digest, filters and every selected accession are recorded in `provenance.json`. Verification at the release: 100 library and 2 integration tests in Rust; Lean 21/21 and 14/14 declarations with zero `sorry` and no unwhitelisted axioms; 33 Python tests; an anti-stub audit over 227 files.

Status of this manuscript. Not peer reviewed. The repository previously described work as "5-round Opus peer reviewed" on the basis of a function that made no model call; that description was withdrawn, and no equivalent claim is made here. Reported results are the output of the commands above, including the three places where that output contradicted what we had written.

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